

HAMOUDA CHEKIR

Junior Software Engineer | Full-Stack & AI

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SUMMARY

Software engineer (ESPRIT, TWIN) focused on delivery: I turn business needs into reliable web products using full-stack development, applied AI and microservices architectures. Comfortable in agile teams and with mixed stakeholders (tech, business, leadership), I prioritise quality, measurability and clean production releases. Recent proof point: at Talan, designed and shipped NextHire, an AI-first ATS of **16 microservices** (React, Node/Express, Python FastAPI, MongoDB, Docker).

EXPERIENCE

Software Engineer — Final-Year Project

Feb 2026 – Jul 2026

Talan / Tunis

- Delivered an AI-first ATS (NextHire) spanning **16 microservices** (1 Node/Express gateway, 2 React SPAs, 13 Python FastAPI/Flask services) by designing a shared MongoDB architecture and Dockerised service boundaries.
- Reduced manual screening effort by automating interview flow: built an adaptive LLM voice agent and a YOLO-based integrity module (face check + prohibited-object detection) for live WebRTC sessions.
- Produced scored hiring decisions for recruiters by implementing a LangGraph pipeline that fused transcription, behaviour and visual signals into ranked candidate reports.
- Accelerated candidate preparation by automating CV parsing (spaCy, PaddleOCR) and LLM-based CV tailoring with anti-hallucination checks, plus adaptive quizzes and Google Calendar scheduling.

Full-Stack Developer Intern

Jul 2025 – Aug 2025

SmartConseil — Dafe Management / Tunis

- Shipped a modular Laravel back office used to manage regions, packs and business modules by rebuilding CRUD workflows in Blade/JavaScript on PostgreSQL and MongoDB.
- Cut fragmentation across crawler tools by migrating HTML/API drivers, grids and IP modules into one centralised Laravel codebase.
- Improved operator efficiency with filterable UIs and PDF/Excel exports; delivered inside an agile cadence (dailies, code reviews, unit tests, SonarQube, GitLab CI/CD, Docker).

Immersion Intern

Jul 2024 – Aug 2024

Big Deal (Club Privilèges) / Tunis

- Enabled digital gift campaigns by designing and deploying a gift-code generation and management module (Laravel, Filament, Git, phpMyAdmin, Postman).

Web Developer Intern

Feb 2023 – May 2023

HOTIX / Tunis

- Launched the “Consommi Tounsi #619” e-commerce site by implementing core storefront features in .NET under Scrum, with test passes and production deployment.

ACADEMIC PROJECTS

Health Record — Django Health Tracking Platform

2025

Django · Python · SQLite · JWT

- Shipped a health-tracking platform (BP, HR, sleep, steps + monthly PDF reports) by combining ResNet50/ResNet18 medical vision (14 thoracic pathologies + brain tumours), recommendations (20+ rules + ML feedback), anomaly scores (Isolation Forest + One-Class SVM, 0–100), a T5 mental-health chatbot and a secured API (JWT, roles/admin, HTTPS).

TuniVert — Environmental Platform

2025

Laravel · PHP 8.3 · MySQL · Stripe/PayPal/Paymee

- Delivered an eco-events and multi-gateway donations platform (Stripe, PayPal, Paymee) with QR verification, gamified challenges, forums and a real-time admin analytics dashboard plus OpenAI donation suggestions/chatbot.

TalentMatch — Secure Auction Platform

2025

Python/FastAPI · React · Qwen2.5-VL · Gemini 2.5 · Serper.dev · Solidity · Blockchain

- Designed a secure auction platform integrating **AI** (authenticity verification, price estimation via web scraping, multimodal listing generation) and **blockchain** (Solidity smart contracts on Ethereum/Sepolia, post-quantum cryptography).
Clinical Decision Support 2025
BLIP · CLIP+LLaMA (LoRA) · RAG · Meditron-7B · Qdrant
- Generated medical reports from radiographs and answered clinical questions with citations by fine-tuning VLMs (BLIP, CLIP+LLaMA LoRA) and a Meditron-7B RAG stack on Qdrant/PubMedBERT, explored with reinforcement learning.
Multi-Agent AML Engine 2025
Business rules · LLM RAG/Critic agents · MCP
- Detected AML patterns (smurfing, loops) by combining business rules, pattern detection and contextual LLM analysis (RAG/Critic agents) behind a secured MCP gateway with access control, immutability and regulatory traceability.

EDUCATION

- Engineering Degree in Computer Science — TWIN** 2023 – 2026
ESPRIT — Private Higher School of Engineering and Technology / Tunis
- Bachelor’s in Business Computing — E-Business** 2020 – 2023
ESEN — Higher School of Digital Economy / Manouba

SKILLS

- Core:** Python, JavaScript, Java, React, Node/Express, FastAPI, Flask, Django, Laravel, MongoDB, PostgreSQL, MySQL, Docker, Git, REST APIs, Scrum/Agile
- AI & Blockchain:** LLM Agents, LangGraph, LangChain, RAG, YOLO, Computer Vision, spaCy, PaddleOCR, PyTorch, scikit-learn, Solidity, Ethereum
- Delivery:** GitLab CI/CD, SonarQube, Postman, WebRTC, RabbitMQ, Keycloak, AWS (Cloud Foundations)
- For this role:** React · Node.js · Python · FastAPI · Microservices · Docker · MongoDB · REST APIs · Agile

CERTIFICATIONS

- CCNA — Switching, Routing & Wireless Essentials** Sept 2024
Cisco Networking Academy
VLAN / inter-VLAN, STP, EtherChannel, WLAN (WLC), L2 security, static IPv4/IPv6 routing
- AWS Academy Graduate — Cloud Foundations** May 2025
AWS Academy · 20h · Credly badge
Compute, storage, networking, IAM, billing, Well-Architected
- Hashgraph Developer Course** Oct 2025
Hedera Hashgraph
Hedera DLT: accounts/keys, HBAR, Consensus/Token services, smart contracts, SDKs
- SDG, Work Ethics & Gender Equity** 2024
Honoris United Universities
UN SDGs, professional ethics, gender equity, transversal soft skills

LANGUAGES

- Arabic** — Native
- French** — Fluent
- English** — Advanced